

Knowledge graph-driven personalized attractions recommendations with tourists' long- and short-term interest modeling

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ABSTRACT

Personalized tourism attraction recommendations are crucial for enhancing travel experiences and supporting the success of tourist attraction development. Despite significant progress in collective wisdom mining and tourist behavior modeling, challenges such as insufficient utilization of data associations and limited interpretability of recommendation results still exist. Knowledge graphs (KGs), with their immense potential, remain fully underexplored in the domain of personalized attraction recommendations. To address this gap, we propose a novel Knowledge Graph Neural Attractions Recommendation (KGNARec) model, which integrates long-term and short-term interests for personalized attraction recommendation. Specifically, we employ Graph Convolutional Networks (GCNs) to encode embedded representations of tourists and attractions, designing a personalized knowledge-aware attention mechanism to distinguish the importance of entities and relationships, thereby uncovering tourists' latent preferences. By recursively aggregating information from multi-hop neighbors within the KG, we can capture high-order structural and semantic information, forming a comprehensive representation of tourists' long-term interests. Simultaneously, we consider tourists' short-term interests, extracting insights from their recent travel records through a personalized attention-based Gated Recurrent Unit (GRU) network. These long-term and short-term interests are adaptively fused through a gated fusion network and then compared with candidate attractions to make personalized recommendations. Experiments on a real-world Beijing tourism KG demonstrate that KGNARec achieves an 8.6% improvement in AUC and a 7.1% improvement in ACC over state-of-the-art methods, highlighting its superior performance and interpretability. These findings provide valuable insights for advancing personalized travel services and the development of smart tourism.

1. Introduction

Tourism is an important part of the contemporary service sector and has become the preferred means for an expanding number of people to satisfy their desires for leisure, recreation, and experiences (Hamid et al., 2021; Law, Li, Fong, & Han, 2019; Li, Hu, Huang, & Duan, 2017). As society enters the post-modern era of tourism, there has been a profound transformation in the behavior patterns and consumer psychology of tourists. Traditional group tours, fixed itineraries, and package services are no longer the top choices for travelers, giving way to a burgeoning predilection for customized and distinctive travel experiences. This shift not only reflects the enhancement of individual autonomy but also signifies the evolution of the tourism market towards greater segmentation and personalization (Chen, Cao, Wang, Liang, &

Zhu, 2022; Gao et al., 2024; Zheng, Liao, & Qin, 2017).

The rapid development of internet technology has provided unprecedented convenience for tourists. The ubiquity of information has transformed social interaction, enabling tourists to easily retrieve a vast array of tourism resources online (Henar, Moya-Gomez, Carlos, & Gutierrez, 2018; Sang, Xu, Qian, & Wu, 2021). While the ease of information retrieval is laudable, it has also introduced the challenge of information overload. Tourism Recommendation Systems (TRS) have emerged as powerful tools to reduce information overload and assist tourists in decision-making (Abbasi-Moud, Vahdat-Nejad, & Sadri, 2021; Zhu et al., 2023). These systems use intelligent algorithms to analyze users' historical behaviors, preferences, and tourism product attributes, recommending travel options that users may find appealing. They play a crucial role in enhancing tourists' travel experiences,

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advancing smart tourism development, and building location-based services (Hamid et al., 2021; Zhang, Li, Li, Wang, & Ding, 2024).

Tourists' travel behaviors exhibit complexity and diversity, encompassing long-term preferences (e.g., sustained interest in specific types of attractions) and short-term interests (e.g., interest driven by specific events or weather). Capturing and balancing these interests poses significant challenges for TRS (Liu, Chen, Huang, Li, & Min, 2023; Wu, Li, Zhao, & Qian, 2022). Traditional recommendation methods, such as Collaborative filtering (CF) and Content-Based (CB) approaches, primarily rely on group similarity and feature extraction to generate recommendations (Jannach, Manzoor, Cai, & Chen, 2022; Sun, Fan, Bakillah, & Zipf, 2015). The recommendation results are relatively transparent, and the algorithm is easy to implement. However, their effectiveness in a big data environment is hindered by data sparsity and the complexity of feature engineering (Chaudhari & Thakkar, 2020; Xu, Chen, Dai, & Chen, 2017). Moreover, these methods mainly adopt a static approach to analyze interactions between tourists and attractions, failing to capture the evolution of users' short-term interests (Li, Hu, & Wang, 2022; Wu et al., 2022). Deep learning techniques excel at automatically extracting high-dimensional features from large-scale datasets and capturing complex user behavior patterns through advanced representation learning (Chen, Cao, Zhu, Wang, & Liang, 2021; Zhu et al., 2021). Existing studies primarily focus on modeling tourists' travel transition patterns to capture short-term interests (Liu et al., 2023; Wang, Yuan, Yang, Peng, & Jiang, 2024). However, these models often transform input data into independent feature variables, neglecting the potential interrelationships and contextual dependencies within tourism data. This approach fails to comprehensively reflect the diversity and complexity of tourists' long-term interests. Additionally, the "black-box" nature of deep learning models and the opacity of their decision-making processes makes the recommendations hard to interpret, reducing their credibility (Batmaz, Yurekli, Bilge, & Kaleli, 2019; Gao, Peng, Lu, Claramunt, & Xu, 2023).

Knowledge Graph (KG) is a new form of knowledge representation, storage, and management, that aims to describe various entities, attributes, and their connections in the real world (Aghaei, Raad, & Fensel, 2022; Chicaiza & Valdiviezo-Diaz, 2021). Large-scale structured triples in KG can integrate various tourism data, uncovering potential preferences aligned with tourists' long-term interests, such as a sustained preference for historical architectural attractions. This integration significantly strengthens the data foundation of TRS (Guo et al., 2020; Ji, Pan, Cambria, Marttinen, & Yu, 2022). The complex relational network enables the algorithm to discover potential connections among attractions, further enhancing the context-awareness capability of the recommendation model and improving the interpretability and diversity of the results (Cai, Xie, Tian, & Cui, 2022; Gao et al., 2023). Current KG-based recommendation methods primarily focus on semantic embedding and path expansion to optimize recommendation performance (Guo et al., 2020; Wang et al., 2024). However, these methods exhibit a narrow focus on structure and semantics, being limited to local path exploration or explicit relationship mining, and fail to fully capture the global structural features and semantic associations within the KG (Li, Ma, & Chu, 2024; Wang et al., 2023). Graph Neural Network (GNN)-based methods frequently overlook the direct modeling of user interests through the KG (Ji et al., 2022; Li et al., 2024; Wang et al., 2024). Since the contextual knowledge from the KG does not apply universally to all user interests. Given that users may have completely different interests in the same entity, selective knowledge absorption is crucial but frequently overlooked in existing research (Wang et al., 2019; Wang et al., 2024; Yang & Dong, 2020). Current tourism recommendation systems often produce a single vector to represent tourist interests, neglecting the need for unified modeling of both long-term and short-term interests (Li et al., 2022; Wu et al., 2022; Zhu et al., 2021). Therefore, effectively utilizing KG to achieve personalized tourist interest modeling and improve the performance of TRS remains a significant challenge.

To address these challenges, this paper proposes a Knowledge Graph Neural Attractions Recommendation (KGNARec) model, a novel framework encapsulating both long- and short-term interests of tourists for personalized attraction recommendation. Firstly, Graph Convolutional Networks (GCNs) encode KG's network structural and semantic information with bias, learning tourists' long-term interests. Entity representations can accumulate messages from their high-order neighboring structures and adjust propagation directions by identifying relationships and entities that reflect potential tourist interests. Subsequently, a Gated Recurrent Unit (GRU) network, enhanced by a personalized attention mechanism, discerns short-term interests from recent travel trajectories. A gated fusion network then harmonizes these two interest representations, producing a comprehensive tourist interest vector. Comparing the final interest vector with candidate attraction vector to predict tourist proclivities visiting specific attractions. To validate the effectiveness of KGNARec, we constructed a tourism KG of Beijing using online repositories and conducted a comparative experiment using travel notes. Empirical findings highlight KGNARec's superiority over state-of-the-art baselines in top-k recommendation and click-through rate prognostications. The KGNARec framework adopts a knowledge-aware modeling approach, achieving high precision and interpretability, and laying a theoretical foundation for smart tourism studies.

Our main contributions can be summarized as follows:

- 1 We explored the application of KG in personalized attraction recommendations by utilizing multi-layer GCNs and personalized knowledge-aware attention to selectively encode semantic information and high-order connectivity, effectively addressing the issue of data sparsity in TRS.
- 2 Long-term interests are derived from the KG, while short-term interests are extracted from travel trajectories using a personalized attention-based GRU network. These interests are then adaptively integrated to construct a comprehensive representation of user interests.
- 3 Empirical validation on the Beijing tourism KG confirms that KGNARec outperforms existing baselines, highlighting its ability to provide interpretable insights and effectively support tourist decision-making.

The remainder of this paper is organized as follows: Section 2 reviews the related work on tourism recommendations. Section 3 presents the background and defines the problem addressed in this paper. Section 4 describes the proposed model. Section 5 exhibits the experimental results, and Section 6 discusses the model's performance. Finally, Section 7 concludes the paper and outlines future research directions.

2. Related work

This section presents a comprehensive review of existing literature on personalized travel recommendation systems, focusing on traditional recommendation methods, deep learning-based approaches, and KG-based techniques, which are highly relevant to and have significantly informed our research.

2.1. Traditional recommendation methods

Collaborative filtering, which makes recommendations by analyzing similar travel behaviors among tourists, has become one of the most successful recommendation techniques (Cui, Luo, & Wang, 2018; Jiang, Qian, Shen, Fu, & Mei, 2015). The core of this method is the collection and analysis of vast amounts of tourist behavior data, providing satisfactory results when sufficient data is available. For example, matrix factorization models decompose the tourist-attraction interaction matrix into two latent matrices that represent the characteristics of tourists and attractions, which can reveal the general preferences of tourists (Cai, Xu,

Liu, & Pei, 2018). However, in the tourism domain, tourists often rate only a few attractions, leading to sparse interactions that hinder finding similar users. Content-based techniques generate recommendations by emphasizing the content features of tourists and attractions, which typically include descriptions, types, locations, and tourist evaluations and ratings of attractions. Feature extraction, a critical step in CB methods, often employs model-driven strategies like Naive Bayes or Decision Trees (Huang & Bian, 2009; Wan, Hong, Huang, Peng, & Li, 2018). Although these techniques can mitigate data sparsity to some extent, their reliance on information extraction poses challenges with unstructured data (Chaudhari & Thakkar, 2020; Xiaoyu Sun, 2018). To address data sparsity, auxiliary data such as ticket prices, geographical locations, and popularity is used to enrich the semantic representation of attractions (Wu, He, Wang, Zhang, & Wang, 2023). However, current methods transform only basic descriptive features like IDs and attributes of tourists or attractions into discrete feature vectors. This approach tends to overlook complex interactions among factors in tourists' decision-making, leading to underutilization of auxiliary information and suboptimal outcomes (Guo et al., 2020; Renjith, Sreekumar & Jathavedan, 2020; Wang et al., 2019).

2.2. Deep learning-based recommendation methods

Deep learning, renowned for its powerful representational learning capabilities, has thriving in the field of contemporary recommendation technologies (Batmaz et al., 2019; Shi, Liu, An, & Chen, 2021; Zhang, Yao, Sun, & Tay, 2019). In the realm of tourist attraction recommendations, these models adeptly navigate challenges like data incompleteness and uncertainty by synthesizing low-level features of tourists and attractions into more abstract, sophisticated high-level features (May Petry, Leite Da Silva, Esuli, Renso, & Bogorny, 2020; Mou et al., 2022; Yi, Zhou, Linna, Yaoli, & Yu, 2020). Given the visual-centric nature of tourism activities, Convolutional Neural Networks (CNNs) are utilized to extract image features of destinations, integrating visual cues into the recommendation model (Ameen, Chen, Xu, Lyu, & Shi, 2020; Chen, Lyu, Yu, & Chen, 2023; Zhang et al., 2024). Users' recent visit records directly reflect their behavioral motivations, needs, and focal points at a given time, effectively capturing their short-term interests. Recurrent Neural Networks (RNNs) and their variants, Long Short-Term Memory (LSTM) and GRU, have demonstrated excellent performance in mobility prediction and can efficiently achieve next-location prediction (Liu et al., 2023; Sperlí, 2021; Wang et al., 2024). To address the issue of data sparsity, researchers have incorporated various contextual information into RNNs. For example, temporal transition matrices and distance transition matrices have been applied to RNNs to mitigate their limitations in capturing sequential information (He et al., 2024; Wang et al., 2024; Yao et al., 2023). While sequence-based deep learning models excel in capturing tourists' short-term interests, they face significant challenges in modeling long-term interests (Cao, Cui & Joe, 2023; Wu et al., 2022). Even when attention mechanisms are introduced to enhance a model's memory capacity, simple weighted calculations often fail to adequately represent users' shifting focus across multiple areas of interest. The introduction of deep learning has greatly contributed to the development of travel recommendation systems in the direction of intelligence (Hamid et al., 2021). However, a notable drawback of these models is their "black-box" problem, which fails to elucidate the decision logic behind tourists' choices in the recommendation stage, potentially undermining the trustworthiness of the recommendations. Moreover, the validity of the model is susceptible to complex network topologies and faces issues such as overfitting (Cai et al., 2022; Gao et al., 2023).

2.3. Knowledge graph-based recommendation methods

The essence of using KGs to enhance recommendations lies in effectively extracting relevant side information and potential

connections (Gao, Gao, Li, Yao, & Chen, 2022; Guo et al., 2020). Path-based strategies introduce the entity and relationship *meta*-paths within KG to identify meaningful connections between users and items (Khan, Ma, Ullah, & Polat, 2022; Li, Sun, Zhang, & Zhang, 2021). For example, Hu, Shi, Zhao, & Yu, (2018) harness priority sampling mechanisms to cherry-pick premium path instances, facilitating the learning of user and item connectivity. Xie, Zheng, Chen, and Zheng, (2021) advocate the use of *meta*-graphs for feature extraction, replacing conventional *meta*-paths. However, the application of such connection patterns requires manual extraction and can only represent linear and simple relationships, which may inadvertently leave out critical information (Cai et al., 2022; Sang et al., 2021; Xie et al., 2021).

Knowledge Graph Embedding (KGE)-based methods use embedding techniques to learn semantic representations of entities and relationships to enhance recommendation performance (Mezni, Benslimane & Bellatreche, 2021; Sang et al., 2021). For example, CKE (Zhang, Yuan, Lian, Xie, & Ma, 2016) extracts semantic information from structured knowledge, textual data, and visual content to enrich the embeddings of users and items. Similarly, Shokrzadeh et al. (2024) utilized the learned embedding representations to enhance the performance of the neural collaborative filtering framework. This approach excels in semantic learning for large-scale knowledge bases, but it may overlook the high-order connectivity inherent in KG, making it more suitable for internal tasks such as KG completion and link prediction (Chen, Jia & Xiang, 2020; Liu et al., 2023; Rossi, Barbosa, Firmani, Matinata, & Meriardo, 2021). While these two methodologies, when considered individually, often focus on isolated aspects of KG. The unified learning framework RippleNet successfully integrates these methods, achieving superior recommendation performance (Gao et al., 2023; Wang et al., 2019).

With the rapid rise of GNNs in information dissemination and graph structure data mining, the motivation to leverage GNNs for KG modeling has grown significantly (Hu, Li, Shi, Yang, & Shao, 2020; Wu, Sun, Zhang, Xie, & Cui, 2023; Zhang, Liu, Zhou, & Chu, 2021). Representative models, such as KGCN, leverage GCNs to extract knowledge relevant to user interests (Wang, Zhao, Xie, Li, & Guo, 2019). Liu et al. (2023) employed a graph attention network to capture users' personalized preferences for specific entities. Wang et al. (2023) designed a deep GCN network with residual links to enhance its ability to capture high-order signals. Wang et al. (2024) developed a lightweight GNN model to capture user preferences for personalized knowledge-aware recommendations by encoding relational paths and associated entities. In the domain of tourism recommendation, Gao et al. (2023) utilized a relational graph convolutional network to extract embeddings of tourist attractions, enabling interpretable recommendations. Zhang et al. (2024) constructed a multimodal tourism KG and employed GCNs to extract features for mining tourists' preferences. These studies demonstrate the significant advantages of GCNs in KG modeling, inspiring us to further explore.

2.4. Summary

Unlike previous studies, this research focuses on integrating tourists' long-term and short-term interests to overcome the limitations of a single-perspective approach. We employed GCNs to extract tourists' long-term preferences and proposed a personalized knowledge-aware attention mechanism that emphasizes the heterogeneity of the KG, enhancing the expressiveness of user preferences. Additionally, a GRU network, enhanced with personalized attention, is developed to capture tourists' short-term behaviors from their travel trajectories. By adaptively integrating these two types of interests, our approach aims to provide accurate predictions and interpretable recommendations.

3. Problem formulation

Before going into details of our approach for personalized travel attractions recommendation, we first introduce some fundamental

definitions and concepts in this paper.

Definition 1 (Tourist-attraction interaction matrix)

Let $U = \{u_1, u_2, \dots, u_m\}$ and $V = \{v_1, v_2, \dots, v_n\}$ denote the set of tourists and attractions, m and n represent the number of tourists and attractions, respectively. The tourist-attraction interaction matrix $Y = \{y_{uv} | u \in U, v \in V\}$ is defined according to tourists' implicit feedback. If tourist u has visited attraction v , then $y_{uv} = 1$; otherwise, $y_{uv} = 0$.

Definition 2 (Tourist visited trajectory)

From the tourists' travel history with timestamps, the attraction visited trajectory $T = \{v_{u,1}, v_{u,2}, \dots, v_{u,i}\}$ of a specific tourist u can be collected, where $v_{u,i}$ indicates the i -th attraction visited by tourist u .

Definition 3 (Tourism knowledge graph)

A tourism knowledge graph G consists of many triplets (h, r, t) , where $h, t \in \mathcal{E}, r \in \mathcal{R}$. Here, $\mathcal{R}$ and $\mathcal{E}$ represent the sets of relations and entities, respectively. As depicted in Fig. 1, the tourism knowledge graph abstracts and links knowledge elements within the tourism domain, forming a structured knowledge graph. Its contents encompass tourism destination information, attraction attributes and tourism route planning, among others.

Formally, given the tourist-attraction interaction matrix Y , the tourism knowledge graph G , and the tourist's recent visited attraction trajectory T , the goal of personalized travel attraction recommendations is to suggest attraction v that may interest a tourists u . To achieve this goal, we develop a predictive function $\hat{y}_{uv} = f(u, v | Y, G, T, \Theta)$, where $\hat{y}_{uv}$ represents the probability of tourist u visiting the attraction v , and Θ denotes the parameter set of the function f . We derive the tourist's long-term interests from the tourism knowledge graph G , and their short-term interests from the recently visited sequence T . We then fuse these interests to better predict the tourist's potential interest in future attractions.

4. Methodology

This section details the methodology of the proposed model, starting with an overview of its overall architecture and followed by in-depth descriptions of its specific components in the subsequent subsections.

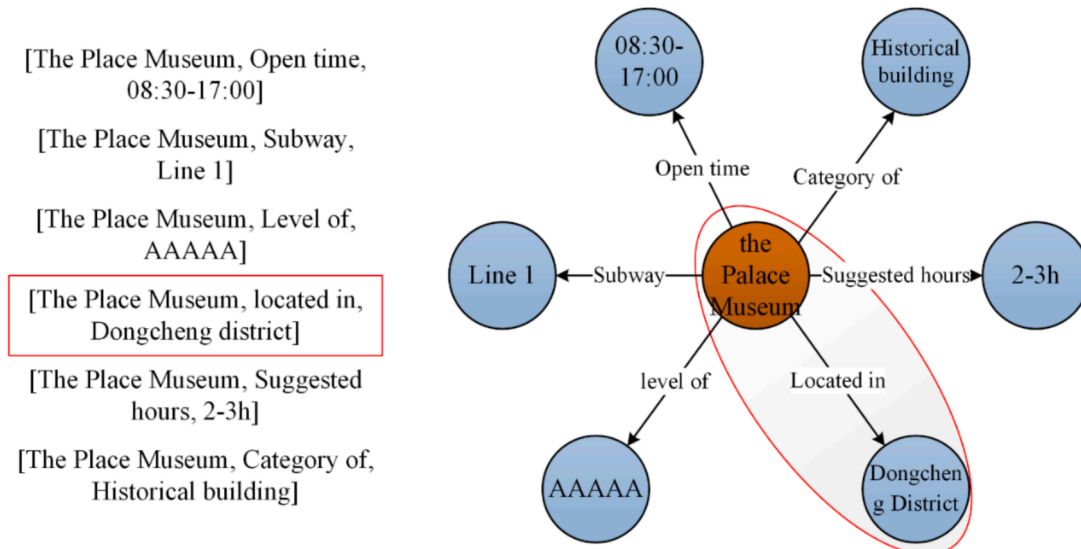


Fig. 1. Illustration of the tourism knowledge graph.

4.1. Overall framework

In this article, we introduce a novel personalized attraction recommendation approach known as the KGNAREc model. As illustrated in Fig. 2, the model consists of two key components: a long-term interest extracting module based on tourism KG analytics and a short-term behavior modeling module that simulate transfer patterns from travel trajectories. The long-term interest module uses an interest propagation method across the KG, leveraging an attention mechanism to effectively propagate tourist preference signals. GCNs are used to encode higher-order connectivity patterns from the KG to represent tourists' long-term interests by deeply mining available contextual metadata. Meanwhile, the short-term behavior module employs a GRU to encode the sequential dynamics of travel trajectories. A personalized attention mechanism is introduced to highlight significant short-term behaviors. The two types of interests are then integrated using a gating fusion method to derive mixed preference scores, which are input into a recommendation layer to assess the likelihood of tourists visiting specific attractions.

4.2. Long-term interest modeling

Long-term interests reflect a tourist's sustained and consistent preferences, which are perceived to remain stable over time and are manifested throughout a tourist's comprehensive travel history (Li et al., 2022; Wu et al., 2022). These interests typically encompass complex semantic information and diverse preference features, which traditional models relying on direct interactions struggle to effectively capture. The tourism KG serves as a comprehensive repository of information and content about tourist attractions. It implies characteristics of attractions that tourists might favor. By associating tourists' historical behaviors with tourism KG, we can quantify the level of interest tourists have towards various attractions, thereby accurately characterizing their long-term interests.

GCN excels at encoding the structural information of graphs, making it an ideal choice for capturing long-term preferences from the KG. By learning node features and modeling relationships, GCN efficiently expresses multidimensional interest, avoiding the heavy reliance on feature engineering typical of traditional methods, producing more expressive and adaptive outputs. Moreover, by recursively aggregating multi-hop neighbor information, GCN effectively uncovers complex semantics and high-order relationships, extracting deeper insights aligned with user interests. This addresses the challenges traditional methods

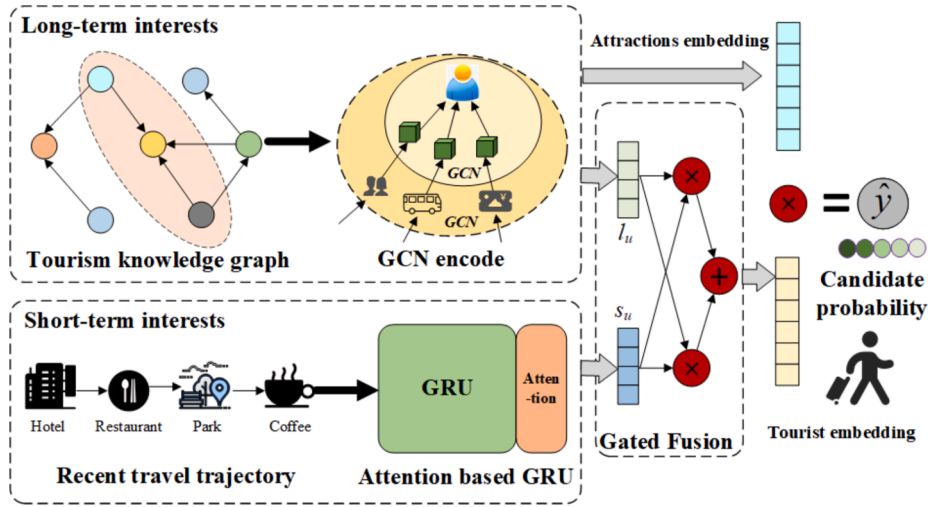


Fig. 2. The overall architecture of the proposed attractions recommendation model.

face in modeling interest connections, particularly when dealing with sparse data and isolated nodes. Thus, like other studies (Hu et al., 2020; Liu et al., 2023), this study employs GCNs to explore long-term user interests.

Specifically, given a tourist-attraction pair (u, v) , let $\mathcal{N}_{e_v}$ denote the neighbor entity set of the entity e_v corresponding to attraction v in the KG, the embedded representation of attraction v encoded by GCN is formally expressed as Eq. (1).

$$v_{GCN} = \text{aggregate}(e_v, \mathcal{N}_{e_v}) \quad (1)$$

Where *aggregate* is the aggregator function, which aggregates information from neighboring entities. In this paper, we use the sum aggregator to take the summation of two representation vectors.

The completeness and efficacy of a KG largely depend on the target entity, its adjacent entities, and the relationships between them. Aggregating all adjacent entities without distinction may introduce unnecessary noise, resulting in imprecise outcomes in practical applications. To overcome this challenge, we utilized a personalized attention module that assigns unique weights to each adjacent entity. During the aggregation process, both the tourist's interest regarding different relationships and the similarity between entities are taken into

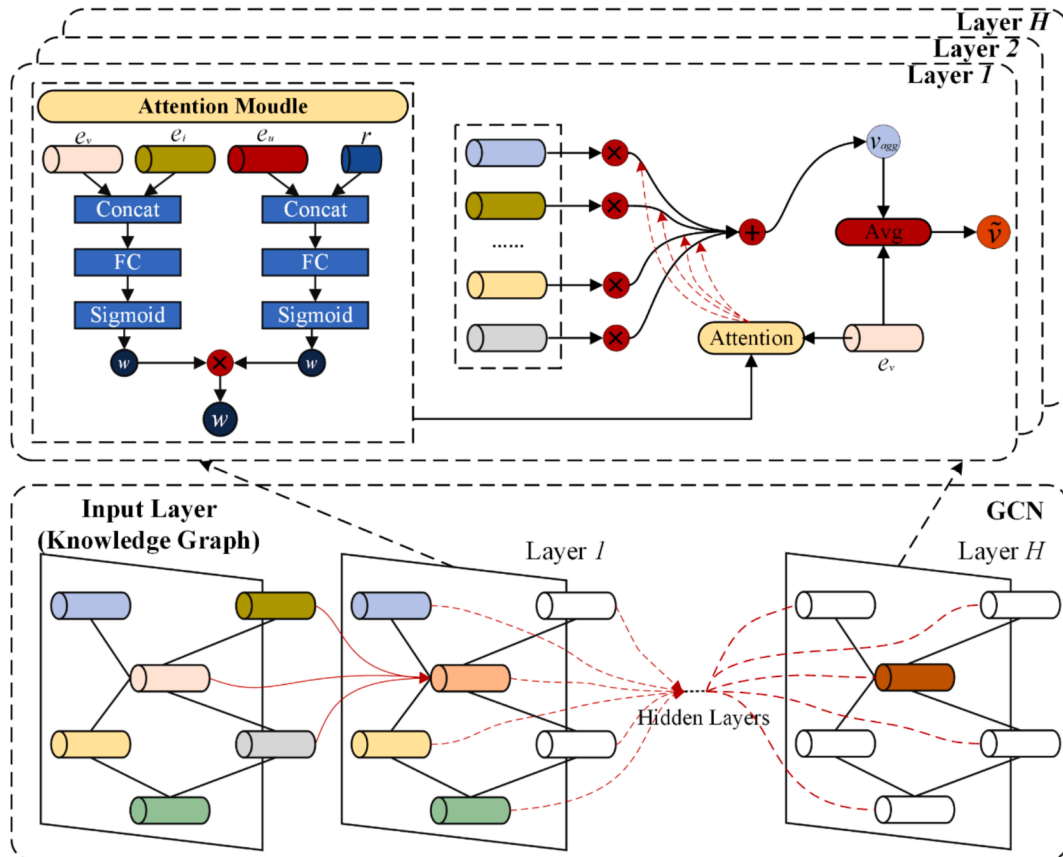


Fig. 3. Illustration of the information aggregation process for GCN.

consideration. For example, when evaluating relationships, some tourists may have a stronger preference for specific types of attractions, while others may prioritize ticket prices. Our attention mechanism acts as a personalized filter, allowing tourists' preferences to propagate purposefully within the KG's structure. As a result, the encoding of tourism information in the KG becomes more efficient and expressive. The computational process is outlined in Fig. 3, with the corresponding formulae presented in Eq. (2), Eq. (3), and Eq. (4), respectively.

$$w_{(e_v, e_i)} = \sigma(w_1 \cdot \text{concat}(e_v, e_i) + b_1) \quad (2)$$

$$w_{(u, r)} = \sigma(w_2 \cdot \text{concat}(e_u, r_{(e_v, e_i)}) + b_2) \quad (3)$$

$$w_u = w_{(u, r)} \cdot w_{(e_v, e_i)} \quad (4)$$

Where $w_{(e_v, e_i)}$ represents the similarity between the attraction e_v and its neighbor e_i . $w_{(u, r)}$ represents the tourist's attention to $r_{(e_v, e_i)}$. w_1 and w_2 are transformation weight, b_1 and b_2 are transformation bias, σ is the nonlinear function softmax.

Then, we apply a cumulative operation on the weighted neighbors to generate an aggregated vector v_{agg} . The final step is to calculate the summation of the original entity representation and neighborhood representation and update the entity representation of v to $\tilde{v}$. The update process of v is shown in Eq. (5) and Eq. (6), respectively.

$$\tilde{v} = \text{aggregate}(e_v, \mathcal{N}_{e_v}) = \text{Relu}(\mathbf{W} \cdot (e_v + v_{\text{agg}}) + b) \quad (5)$$

$$v_{\text{agg}} = \sum_{e_i \in \mathcal{N}_{e_v}} e_i \cdot w_u \quad (6)$$

where $\mathbf{W}$ and b represent the transformation weight and bias, respectively. It is important to note that the size of the adjacent entity set within the tourism KG may exhibit significant variations. To ensure computational efficiency and maintain a consistent calculation approach, this research adopts a uniform sampling strategy in which K adjacent entities are uniformly selected from the target entity, disregarding the complete neighborhood. This approach streamlines the computational process and helps to eliminate unrelated entities and reduce noise.

The process pertains to a single-layer GCN operation, where the representation of the target entity relies solely on itself and its adjacent neighbors. It is possible to naturally extend the sampling depth of GCN from a single layer to multiple layers. During each iteration, the representation of each entity and its neighborhood is aggregated to serve as the foundation for subsequent propagation. As the iteration depth increases, the target entity progressively accumulates more extensive and deeper neighborhood information, thereby enabling a more precise exploration of the latent interests of tourists. After the propagation of the final layer, the h -th layer embedded vector $\tilde{v}^{(h)}$ of the candidate attractions v is obtained, which integrate higher-order neighborhood information.

After H aggregation layers, a series of attractions embeddings $v = \{\tilde{v}^{(1)}, \tilde{v}^{(2)}, \dots, \tilde{v}^{(H)}\}$ is obtained. The outputs of different layers retain connectivity information at various orders. Thus, a linear layer-aggregation mechanism is employed to fuse the embedding representations from all layers, producing the final attraction embedding: $v^* = \sum_{h=0}^H \frac{1}{H+1} \tilde{v}^{(h)}$. By further aggregating the embedded representations of attractions that the tourist has historically visited, we can derive the long-term interest vector u_l of the tourist u . The equation can be expressed as Eq. (7).

$$u_l = \text{GCN}(u, v_i^*) \quad (7)$$

This tourist embedding, based on a complete travel history, effectively captures tourists' long-term interests. However, we also recognize that tourists may exhibit a propensity towards engaging in transient activities. Therefore, short-term interests should also be considered in

personalized attraction recommendations.

4.3. Short-term interest modeling

Short-term interest refers to unstable preferences triggered by certain conditions and stimuli in the current environment, which have real-time impact on the tourist's current choices. This preference is mainly manifested in recent historical visits. The travel trajectory of tourists exhibits typical time-series characteristics, and RNNs are the mainstream deep learning models for modeling sequence relationships. Therefore, in this study, we adopt the GRU as the implementation of the RNN and incorporate a personalized attention mechanism to measure the weights of various attractions in the trajectory sequence, aiming to optimize the learning ability of short-term interests. The network is illustrated in Fig. 4.

The GRU network, with its distinctive gate unit design, exhibits a high sensitivity to temporal information and offers advantages in addressing issues related to long-term memory. As shown in Fig. 4, the GRU network takes the tourist's recently visited attraction v_i as input. It uses the hidden state vector h_i , along with the previous hidden states $\{h_1, h_2, \dots, h_{i-1}\}$, to encode each interaction record, resulting in the sequential feature representation of the tourist. As detailed in Eq. (8).

$$h_i = \text{GRU}(h_{i-1}, v_i) \quad (8)$$

The current travel decisions of tourists are influenced by their past behaviors and by identifying important visits in their travel history, we can understand their current travel motivations. In conventional non-personalized attention networks, the attention weights are calculated based solely on a fixed attention vector, which does not capture the variations in preferences among tourists. Therefore, this study proposes the use of a personalized attention network to capture the relevance of the same attraction to individual tourists, which allows for the identification of significant travel behaviors based on individual preferences. As shown in Eq. (9) and Eq. (10), the personalized attention mechanism receives the GRU output h_i and the current tourist embeddings u as input, computing relevance scores a_i for each hidden state h_i in the sequence, which are subsequently normalized. These scores indicate the importance of each travel event in the sequence for predicting the tourist's next action, allowing the model to dynamically focus on key visits.

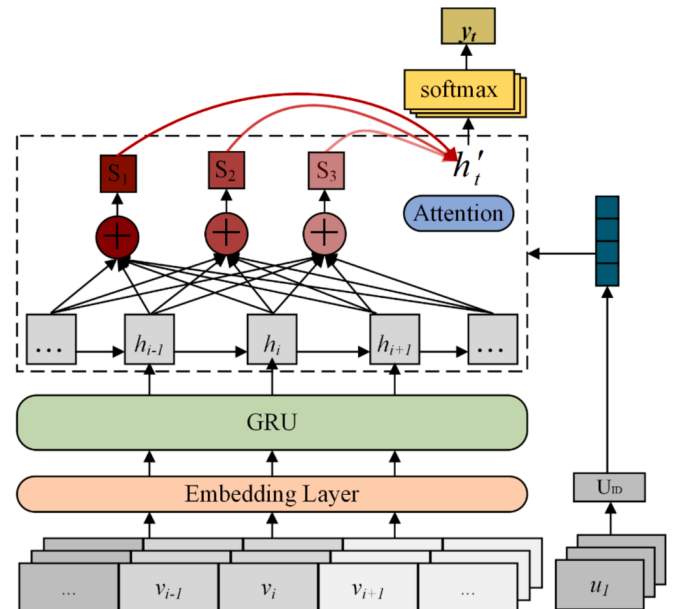


Fig. 4. Short-term interest modeling framework.

$$a_i = \text{utanh}(\mathbf{W}_h \mathbf{h}_i + b_h) \quad (9)$$

$$\alpha_i = \frac{\exp(a_i)}{\sum_j \exp(a_j)} \quad (10)$$

where $\mathbf{W}_h$ and b_h are the parameters. α_i represents the attention weight of the i -th visited attraction in the travel sequence. Finally, based on the attention weights, a weighted sum of the hidden states across all time steps is computed to generate a context vector, as detailed in Formula (11). This vector summarizes the sequential visiting behaviors and primary visiting intentions of the current tourist, capturing their short-term interests s_u .

$$s_u = \sum_i \alpha_i \mathbf{h}_i \quad (11)$$

4.4. Long-term and short-term interest fusion

So far, we have learned two interest representations, the short-term interest s_u and the long-term interest representation l_u . In traditional approach, these two interest representations are typically concatenated linearly to represent the overall interest of the tourist. However, different tourists may rely on long-term and short-term interests to different extents, this study utilizes a personalized gated fusion network instead of the traditional linear combination method. It utilizes learnable gates to balance the personalized weights assigned to long-term and short-term interests. Specifically, given the short-term preference s_u , long-term preference l_u , and tourist embedding u as inputs, the gate vector u_f is computed as shown in Eq. (12).

$$u_f = \text{sigmoid}(\mathbf{W}_q u + \mathbf{W}_s s_u + \mathbf{W}_l l_u + b_u) \quad (12)$$

where $\mathbf{W}_q$, $\mathbf{W}_s$, $\mathbf{W}_l$ and b_u are the projection parameters. The final tourist's interest u^* can be calculated by Eq. (13).

$$u^* = (1 - u_f) \odot s_u + u_f \odot l_u \quad (13)$$

where $\odot$ is element-wise multiplication.

4.5. Optimization

Finally, the final tourist interest representation u^* and the candidate attraction embedding v^* are fed into a deep neural network (DNN) to predict the probability of tourists visiting attraction v . As shown in Eq. (14).

$$\hat{y}_{uv} = \text{DNN}(u^*, v^*) \quad (14)$$

To train our model, we adopted a negative sampling strategy. For each input sample $X = (u, v, y)$, an evaluation probability will be generated after model training, indicating the probability of tourist u will visit candidate attraction v . The cross-entropy loss is used as our loss function L , as shown in Eq. (15).

$$L = - \left\{ \sum_{X \in \Delta^+} y_{uv} \log \hat{y}_{uv} + \sum_{X \in \Delta^-} (1 - y_{uv}) \log (1 - \hat{y}_{uv}) \right\} + \lambda \|\Theta\|_2 \quad (15)$$

where Δ^+ and Δ^- represent the positive sample set and the negative sample set, respectively. The last term is the L2-regularizer and λ is the penalty weight. Θ denotes the parameter set, primarily comprising the learned embeddings in the model and hyperparameters, including weight matrices and biases. Besides, we also apply dropout to avoid over-fitting.

5. Experiments

We executed a series of empirical experiments to assess the effectiveness of the KGNARec method we introduced. The objective was to

answer the following research questions:

RQ1: How well does KGNARec perform? Does it outperform state-of-the-art approaches?

RQ2: How do the key parameters affect the performance of KGNARec?

RQ3: What is the contribution of each critical component of KGNARec to the understanding and modeling of tourist decision-making?

RQ4: Whether KGNARec can unravel the decision-making process behind tourist selection behavior and provide an explanation for its recommendation outcomes.

5.1. Construction of tourism knowledge graph

Using Beijing as a case study, this paper integrates social media data to construct its tourism knowledge graph for the city. First, entities and relationships related to destinations are extracted from CN-DBpedia, which serves as the structured data source for building the tourist knowledge graph. Attributes and values are normalized using the definitions in CN-DBpedia to ensure well-organized knowledge structure. CN-DBpedia, the largest open Chinese knowledge graph, is widely utilized in intelligent question answering and content search. Second, basic information about attractions (such as ticket prices, transportation methods, and opening hours) is extracted from the Information Boxes (InfoBox) of Qunar and Baidu Baike through web parsing and regular expressions. Descriptive information about attractions and tourist activities is also extracted from unstructured data sources like user reviews and travel notes, using BERT-BiLSTM-CRF technology. This enriches and complements the tourist knowledge. Qunar.com (<https://www.qunar.com/>), China's leading online travel platform, provides real-time tourism product inquiries and information services to tourists. Qunar's data has been extensively used in studies on destination image perception and tourist mobility analysis (Jin, Cheng & Xu, 2018; Mou et al., 2020; Zheng, Mou, Zhang, Makkonen, & Yang, 2021). Finally, tourism knowledge from various sources is evaluated, integrated, and stored as triples in the Neo4j database.

Based on the above procedure, we have constructed a tourism knowledge graph of Beijing. This knowledge graph contains attribute information of 844 attractions in Beijing, including 6724 entities, 14 relationships, and 23,676 triples. Fig. 5 displays a partial representation of the Beijing tourism knowledge graph.

5.2. Experimental setup

In this section, we describe the experimental setup, including the datasets, baseline methods, parameter configurations, and evaluation metrics used for comparison.

5.2.1. Datasets

To validate the model's effectiveness, we collected 4,630 travelogue datasets from Qunar.com for our study. In the interaction records, we reconstructed travel trajectories based on the visit sequence. An interaction was marked as 1 if the user visited a particular attraction, and 0 otherwise. To maintain data quality, we excluded user trajectories with fewer than three attractions and attractions with less than 10 visits. The resulting dataset contained 19,522 valid interaction records from 3,944 tourists. Fig. 6 shows the distribution of visits to various tourist attractions in Beijing.

5.2.2. Baselines

We compared KGNARec with four groups of methods for travel attractions recommendation.

The traditional methods are as follows:

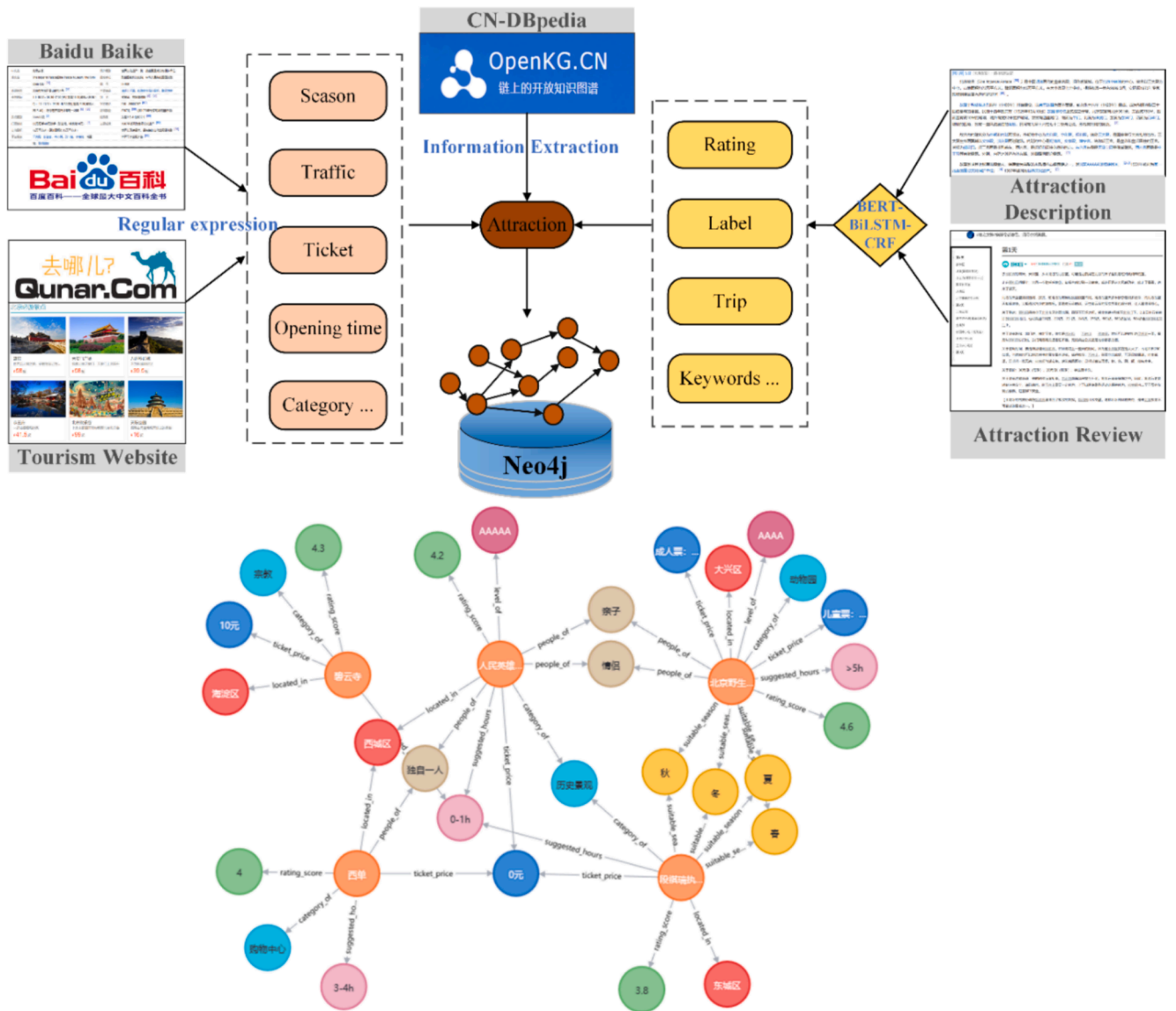


Fig. 5. Part of Beijing tourism knowledge graph.

Item-KNN: An item-based CF approach that recommends attractions like those a tourist has interacted with historically.

GMF: A generalized matrix factorization model that predicts tourist preferences by multiplying the tourist interest matrix with the attraction feature matrix.

The deep learning-based recommendation methods are as follows:

RNN: This is a state-of-the-art temporal prediction method, which has been successfully applied in the next POI prediction.

NCF (He et al., 2017): A neural network architecture for CF, using a multi-layer perceptron instead of inner product computation.

DeepMove (Feng et al., 2018): This method employs an attention mechanism to model the user's long-term interest from historical data, while capturing short-term interest using RNN module trained on the current trajectory.

PLSPL (Wu et al., 2022): A sequence prediction model that employs an attention mechanism to capture long-term preferences and utilizes two LSTM to separately model short-term preferences for location-based and category-based sequences.

The KG-aware recommendation methods are as follows:

CKE (Zhang et al., 2016): A CF recommendation model that integrates KGE technology. This study only utilizes the structured knowledge extracted by TransR as the input.

RippleNet (Wang et al., 2019): A KG-based recommendation model that treats a tourist's historical interests as a seed set within the KG, and iteratively expands the tourist's interests along KG relationships.

Finally, we present the GCN-based recommendation methods:

KGCN (Wang et al., 2019): A recommendation algorithm utilizing GCN for mining attraction features in the KG. When calculating the representation of a given entity, it summarizes relationship information with bias.

COAT (Dai et al., 2022): A state-of-the-art GNN-based recommendation model that simultaneously models user-item graphs and KG using an efficient GCN and a KG-aware attention mechanism, aimed at capturing information-rich collaborative signals and semantic information for enhanced recommendations.

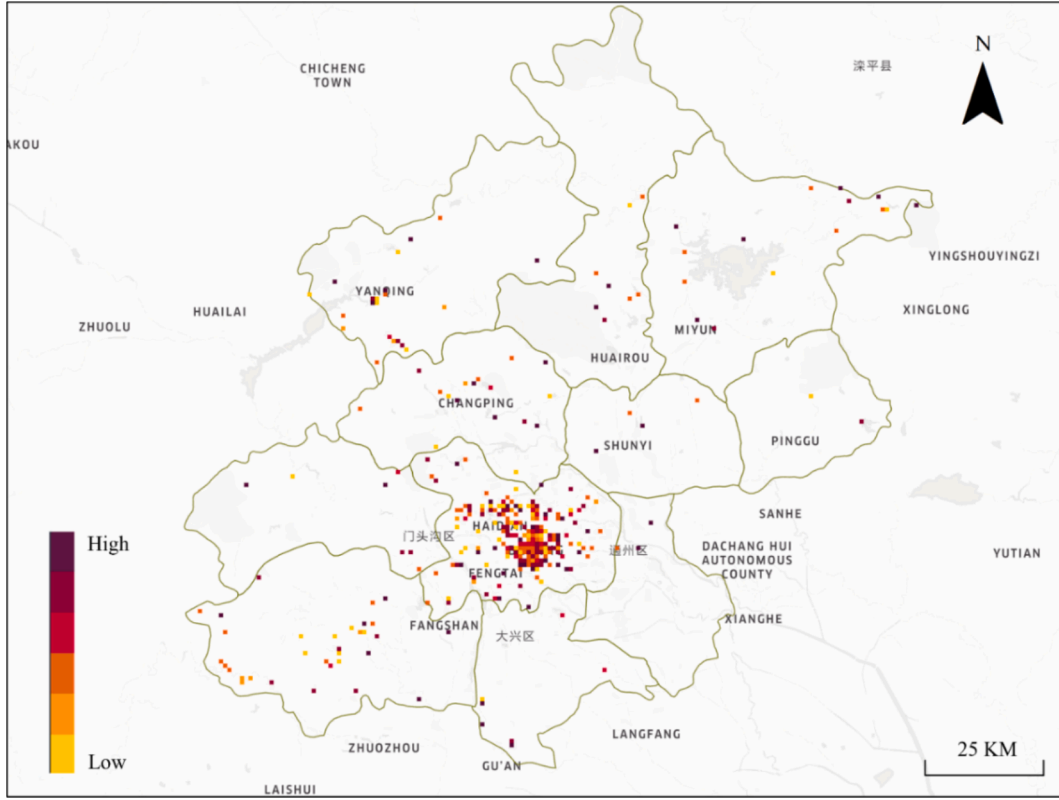


Fig. 6. The spatial distribution of tourist attractions in Beijing.

GLSP (Liu et al., 2023): A latest location prediction model that employs GCNs to encode long-term preferences from location graphs and utilizes an LSTM network to capture short-term preferences from travel trajectories.

5.2.3. Parameters and metrics

In this study, we partitioned the dataset into training and test sets with an 8:2 ratio, reserving 10 % of the training set as a validation set for hyperparameter tuning. Trainable parameters were initialized using the Xavier initialization method, with the Adam optimizer and a batch size of 128. Hyperparameters were selected through grid search strategy. Specifically, the learning rate was varied within $\{0.0005, 0.001, 0.005, 0.01, 0.05, 0.1\}$ and fixed at 0.001; the L2 regularization coefficient within $\{10^{-5}, 10^{-4}, \dots, 10^{-1}\}$ and fixed at 0.005; and dropout values within $\{0.0, 0.1, \dots, 0.5\}$ and fixed at 0.5. For GCN-based models, the number of layers varied within $\{1, 2, 3, 4\}$, and neighbor sampling sizes within $\{2, 4, 8, 16, 32\}$. For baselines, parameters were configured as specified in the respective authors' original paper. An early stopping strategy was employed, terminating training if the Recall metric did not improve for 30 consecutive epochs. All experiments were conducted on consistent hardware, comprising an Intel i9-10900 CPU, 64 GB of RAM, and an NVIDIA RTX 3060 GPU running Windows 10.

We utilize two commonly evaluation metrics in our study. First, for click-through rate (CTR) prediction, the trained model is used to predict the interactions between tourists and attractions in the test set. The evaluation metrics employed are the Area Under the Curve (AUC) and accuracy. Second, for Top-N recommendation, the trained model selects the top N candidate attractions with the highest predicted probabilities for each tourist in the test set. The recommendation results are evaluated using Precision@N, Recall@N, and F1@N ($N = 2, 5, 10, 20, 50$). All experiments were conducted five times, with the average value recorded as the final result.

5.3. Performance comparison (RQ1)

The results are shown in Fig. 7 and Table 1. The key observations are as follows:

Item-KNN relies on similarity calculations between attractions but neglects users' personalized preferences, resulting in the worst performance. GMF improves recommendation by learning latent feature vectors to capture user preferences but remains limited by its linear interaction modeling. NCF replaces traditional inner product calculations with neural networks, outperforming CF methods and demonstrating the capability of deep learning models to capture nonlinear user behavior. RNN-based methods effectively capture tourists' sequential travel patterns, providing valuable insights over short-term periods. DeepMove and PLSPL further leverage attention mechanisms to learn periodic patterns of long-term preferences, outperforming simple RNNs. However, DeepMove focuses solely on periodicity, leading to inferior performance compared to PLSPL, which models both category- and location-based sequences. This highlights the importance of fine-grained modeling of long-term and short-term interests for TRS. However, the utilization of auxiliary information in PLSPL remains isolated and fragmented, resulting in performance inferior to our model.

KG-aware models, such as CKE, RippleNet, and KGCN, have significantly improved recommendation performance, demonstrating the positive role of KG in capturing user preferences and alleviating data sparsity. RippleNet and KGCN leverage multi-hop neighborhood information to further enhance performance, proving that deep semantic information and high-order relationships are crucial for guiding recommendations. COAT introduces a personalized KG-aware attention mechanism that considers user preferences for different relationships and entities, outperforming KGCN. This indicates that information in the KG is not always beneficial. Fine-grained exploration can reduce noise interference and achieve better performance. Although COAT encodes collaborative signals among users and leverages collective intelligence to enhance recommendations, it overlooks the dynamic and short-term

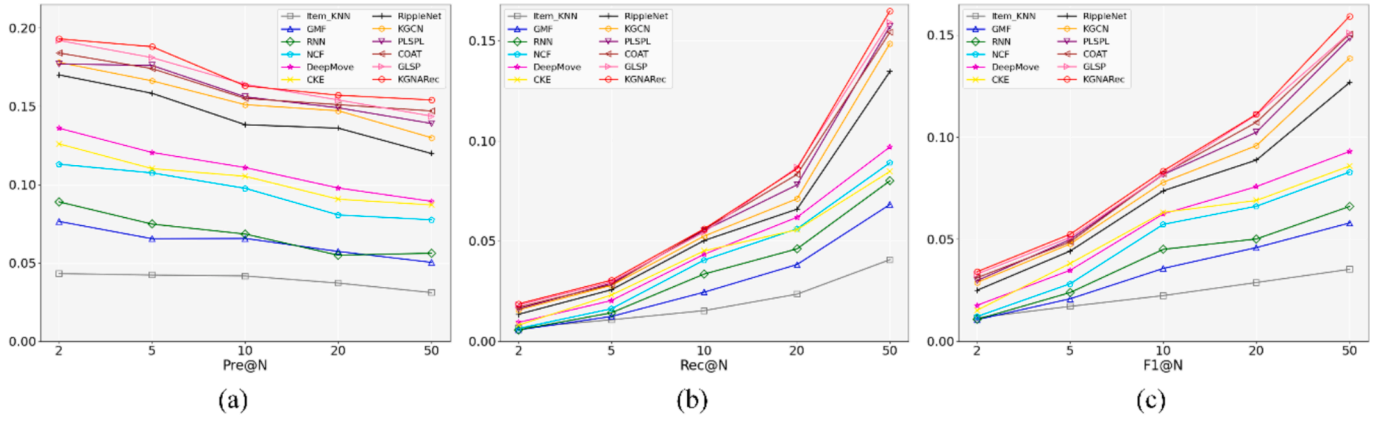


Fig. 7. Recommendation performance of the different models. (a) Precision@N, (b) Recall@N, (c) F1@N.

Table 1
Comparison of Different Models.

	Models	AUC	ACC
Traditional recommendation methods	Item-KNN	0.703	0.648
	GMF	0.738	0.667
Deep learning-based methods	RNN	0.774	0.689
	NCF	0.845	0.732
	DeepMove	0.859	0.772
	PLSPL	0.919	0.839
KG-aware methods	CKE	0.837	0.754
	RippleNet	0.894	0.827
GCN-based methods	KGCRN	0.908	0.843
	COAT	0.917	0.844
	GLSP	0.924	0.849
Ours	KGNARec	0.932	0.855

randomness in tourists' preferences.

GCN-based methods outperform most non-GCN approaches, such as Item-KNN, CKE, and RippleNet, demonstrating the potential of GCN in mining graph-structured data. Compared to PLSPL, which discretely processes location information, GLSP transforms location data into a graph structure to model long-term preferences, achieving superior performance and demonstrating the effectiveness of graph structures in associating tourist preferences. However, spatial relationship graphs contain limited information, leaving many aspects of tourists' interests unaddressed and ultimately resulting in inferior performance compared to our model.

The KGNARec model outperformed all evaluated models, achieving improvements of 8.6 % and 7.1 % over the strongest baseline model in AUC and ACC, respectively. Its exceptional performance primarily attributable to several key factors: (1) The introduction of KG effectively

organizes the knowledge and relationships in tourism data, mitigating the sparsity of tourist-attraction interactions; (2) The model uses GCNs to construct embedded representations from the KG by considering tourists' interests with entities and relationships. This process reveals the long-term patterns in tourists' behavior and enhances the guiding role of the KG; (3) The architectural design of the model has harmonized the short-term preferences with the long-term interests of tourists, enabling a more comprehensive reflection of tourists' latent preferences.

5.4. Effects of parameters (RQ2)

In this section, we investigate the effects of several key hyper-parameters on the performance of the proposed model. These include the neighborhood sampling size K , the number of GCN layers H , and the dimensions of entity-relationship embeddings D (which are set as the same). The results are illustrated in Fig. 8.

Firstly, we examine the impact of the neighborhood sampling size on the model's performance, as shown in Fig. 8 (a). As the number of samples K increases, the model's performance initially exhibits an upward trend. This improvement can be attributed to the fact that a smaller K may fail to capture sufficient neighborhood information, while increasing the sample size can offer a broader spectrum of tourism knowledge. However, as K continues to increase, the model's performance starts to decline, indicating that selecting a smaller, representative subset of neighborhood is already sufficient. Excessive neighborhood sampling not only increases computational overhead but may also introduce noise, leading to converging representations among entities.

Next, we investigate the effects of the number of GCN layers H , as shown in Fig. 8 (b). A small H may limit the model's capability to explore long-distance entity associations, thereby diminishing the effectiveness

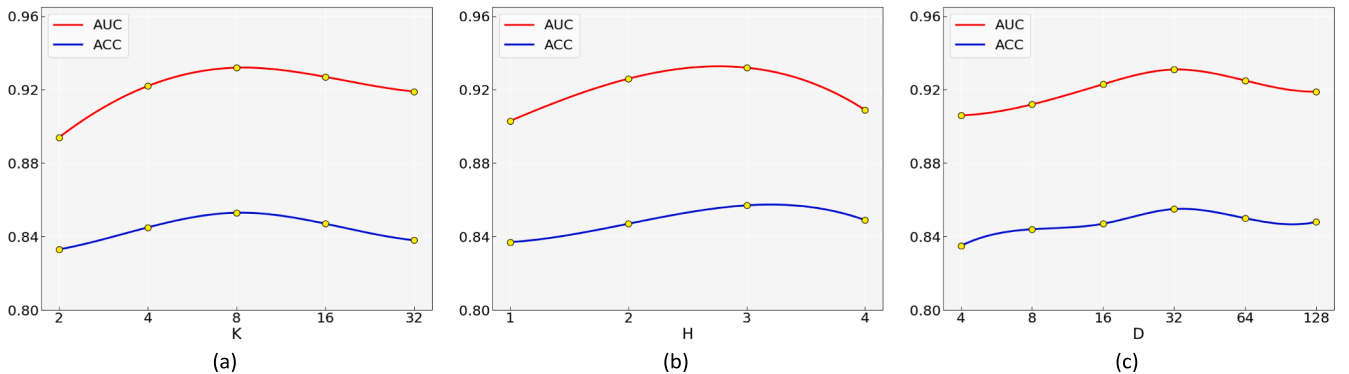


Fig. 8. Parameter sensitivity analysis of KGNARec. (a) Different sampling size, (b) Different layers of GCNs, (c) Different embedding dimensions.

of capturing higher-order relationships. Conversely, an overly large H might result in valuable information being overshadowed by noise. This aligns with our observation that exceedingly long relational chains don't hold practical significance when computing entity similarity. When H is set to 3, the model reaches optimal performance, likely due to a balance between the positive signals from long-distance associations and the negative noise signals.

Lastly, we evaluate the influence of embedding dimensions size, as shown in Fig. 8 (c). Preliminary results indicate that a larger embedding dimension can more adeptly encode semantic information from the KG, thereby enhancing model performance. However, as the dimensionality continues to swell, the performance begins to decline, likely due to difficulties in computing similarity and the model exhibiting overfitting tendencies. When the dimension is set to 32, the model achieves its optimal performance, suggesting that this dimension setting is most adept at capturing the semantic relationships among attractions.

5.5. Model component analysis (RQ3)

To verify the contribution of each component of KGNARec, we implemented four variants of KGNARec for ablation experiments. As shown in Table 2, the first two variants are derived from KGNARec by removing specific modules. KGNARec-neighbor mean aggregation is a simplified version of KGNARec, which directly performs mean aggregation on adjacent entities without considering the deviation of relationships and entities. It aims to evaluate the effectiveness of the designed "attention mechanism". The KGNARec-linear concatenation model adopts a linear concatenation approach to combine long-term and short-term interests, aiming to explore the impact of different interest fusion strategies on model performance.

When focusing solely on the short-term interests of users, this variant performed the worst among all tests, suggesting that short-term interests alone are insufficient for optimal recommendations. Models that focus exclusively on the long-term interests of users significantly outperform those considering only short-term interests. This suggests that for attraction recommendations in our study, long-term consistency is more crucial than short-term randomness for maintaining the overall system performance. Nonetheless, both long-term and short-term interests are indispensable for tourist attraction recommendations. Compared to KGNARec-neighbor mean aggregate, the superior performance of KGNARec underscores the value of a more granular examination of entities and their interrelations within the KG, thereby harnessing its directive potential more effectively. Furthermore, the implementation of the "attention mechanism" within KGNARec adeptly captures the nuances of tourist preferences and steers the flow of interest, mirroring the tourists' primary behavioral patterns. In comparison with KGNARec-linear concatenation, the superior performance of KGNARec suggests that its fusion gate design modulates the contribution of both long-term and short-term tourist interests, improving recommendation efficacy. Collectively, these findings affirm the model's advanced capability in discerning and learning from tourist preferences.

5.6. Interpretability study (RQ4)

To visually demonstrate the interpretability of recommendation models based on the KG, we reveal the decision logic behind tourists'

choice behavior through a case study. As shown in Fig. 9 (a), different node sizes describe different entities' weights. For clarity, we have omitted the names of certain relationships and some unrelated interest paths.

The tourism KG connects candidate attractions to entities with diverse advantages, creating an interest distribution network. Tourists' historical visits are linked to candidate attractions via multiple paths in the KG. These paths, identified automatically through preference propagation, significantly improve the interpretability of recommendation results. The reasoning behind recommending the Temple of Heaven Park exemplifies this well. From tourists' travel records, it is evident that his interests span relationship types like "attraction type", "transportation", and "travel group". Attractions related to these types (e.g., "The Palace Museum", "Qianmen Street", "Tiananmen Square") are closely linked with the Temple of Heaven Park. Furthermore, among the various link paths from tourists' travel histories to candidate attractions, entities like "short weekend trips" and "historical architecture" attract more intensive attention. These frequently visited hub entities can be viewed as potential profiles of tourists and provide key clues for future recommendations. In summary, the KGNARec model significantly enhances the interpretability and transparency of recommendations by leveraging graph-based inference in the KG, underscoring its potential for personalized recommendations.

6. Discussion

Accurate personalized attraction recommendations help to reveal the complexities of human behavior and travel patterns. Traditional methods primarily extract insights from tourists' historical data and collective wisdom. However, these approaches predominantly account for tourists' direct behaviors, often overlooking intricate interconnections inherent in tourism knowledge (Sang et al., 2021; Shokrzadeh et al., 2024; Wang et al., 2019). KG can provide richer contextual information for recommendation systems, effectively alleviating the problem of data sparsity. The fusion of multi-level information can expand the information of the original attractions and improve the accuracy and diversity of recommendations (Gao et al., 2024; Sang et al., 2021). This is the key motivation behind introducing KGs to enhance recommendation performance in existing studies.

A deep understanding and effectively integration of tourists' long-term and short-term interests are crucial for achieving highly personalized services (Chaudhari & Thakkar, 2020; Zhu et al., 2021). This study innovatively introduces KG for tourists' long-term interests mining, providing a stable preference framework for the recommendation system, allowing us to gain insights into tourists' fundamental tendencies and enduring interests. It addresses the limitations in existing studies regarding the insufficient characterization of the complexity and diversity of long-term interests. Simultaneously, considering short-term interests from tourists' travel trajectories introduces necessary flexibility and timeliness to TRS. This means that the system can rapidly respond to tourists' immediate needs and situational changes. Experimental results convincingly demonstrate that KGNARec effectively overcomes the limitations of existing TRS models that rely on single-interest dimension modeling. The balanced integration of long-term and short-term interests effectively considers the diversity of tourists' interests and individual differences (Li et al., 2022; Wang, Li, Zhang, & Wei, 2022; Wang et al., 2024). Compared to simple averaging or direct embedding, this method significantly enhances the expressiveness of user embeddings, ensuring that the TRS remains faithful to tourists' core preferences.

KG-aware recommendation systems represent a critical research direction for promoting interpretable and controllable AI. Compared to traditional GCN models, KGNARec employs a more flexible and refined strategy for information propagation and aggregation. Traditional GCN typically uses uniform weights for neighbor aggregation, which, while simple, may not be precise enough when dealing with complex entities and relationships. KGNARec adaptively assigns different weights to each

Table 2

Ablation analysis of the designed KGNARec.

Models	AUC	ACC
KGNARec-only short-term interest	0.829	0.738
KGNARec-only long-term interest	0.912	0.833
KGNARec-neighbor mean aggregate	0.920	0.837
KGNARec-linear concatenation	0.927	0.842
KGNARec	0.932	0.855

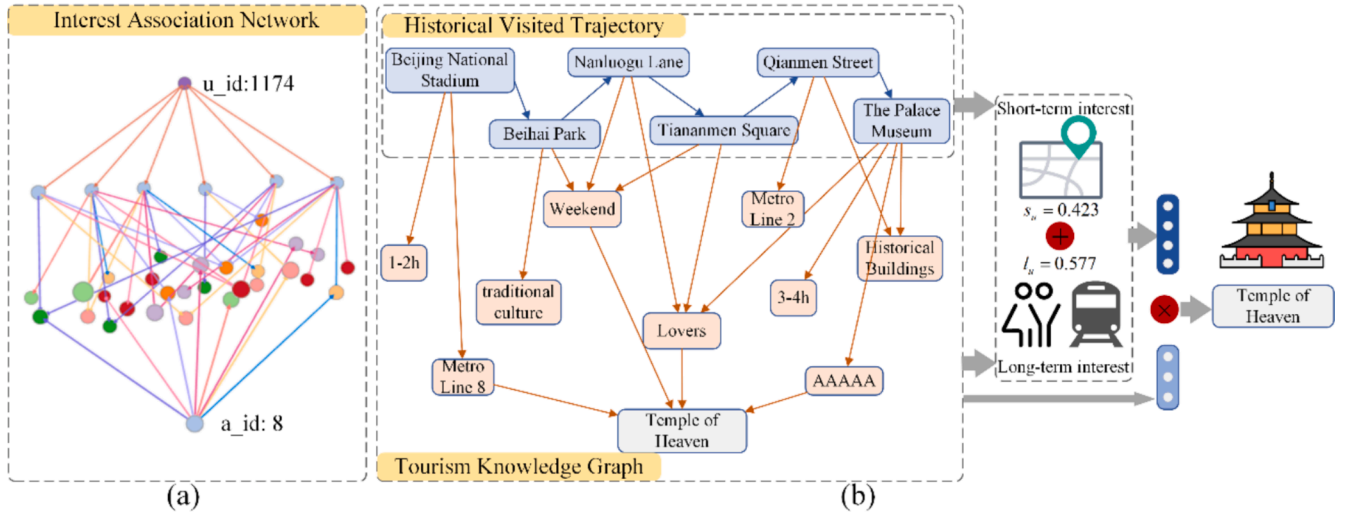


Fig. 9. Case illustrating how KGNARec uncovers the travel decision-making process, including (a) the interest association network in the KG, and (b) visualization of how the KGNARec method works.

neighbor through its customized “attention mechanism”. This strategy allows the model to accurately discern the heterogeneity of entities and relationships in the KG (Guo et al., 2020; Li et al., 2024; Wang et al., 2024), reducing noise and irregularities interference while enhancing the robustness and adaptability of GCNs in practical applications (Gao et al., 2023; Sang et al., 2021). Additionally, the interest propagation framework used in KGNARec can automatically infer the linking paths from travel history to candidate attractions without the need for predefined *meta*-paths or graphs. This further enhances the interpretability of recommendations, allowing the model to better understand and explain tourists’ decision-making processes (Cai et al., 2022; Gao et al., 2024). The KGNARec model comprehensively leverages these advantages, providing robust support for modeling tourist interests and improving recommendation performance.

In practice, our research demonstrates significant application value in enhancing tourists’ travel experiences and strengthening destination management. For travelers, the proposed model can suggest the most suitable attractions and travel activities, saving time on itinerary planning and optimizing the decision-making process. The recommendation process is presented transparently to visitors, greatly enhancing their understanding and trust in the recommendations. This transparency helps build public trust in the recommendation systems, ensuring their fairness and reliability, while addressing concerns about issues such as privacy leakage and the potential misuse of big data (Chaudhari & Thakkar, 2020; Hamid et al., 2021; Wang, Wang & Lu, 2023). For destination management organizations, analyzing and integrating the recommendation results allows them to stay informed about trends in the tourism market. With this information, organizations can refine destination market positioning and product innovation to enhance the visibility and appeal of the region. For example, tourism communities can accurately identify potential customers and implement tailored marketing strategies or location-based advertising services (Deng, Liu, Dai, & Li, 2019; He et al., 2024; Zheng, Huang & Li, 2017). By analyzing real-time visitor data, KGNARec can also predict the temporal and spatial distribution of tourists, enabling real-time monitoring and forecasting of future congestion levels at attractions. For example, when a popular attraction nears its capacity limit, the system can recommend nearby attractions with similar appeal but lower visitor density. This approach not only reduces waiting times and overcrowding but also mitigates risks such as stampedes and infrastructure strain. These outcomes provide data support and decision-making basis for promoting the development and innovation of tourism destinations, achieving more scientific tourism resource management and sustainable

development.

7. Conclusions and future work

In this paper, we introduced KGNARec, a novel knowledge graph neural attraction recommendation model that integrates tourists’ long- and short-term interests for personalized travel recommendation. The model innovatively leverages GCNs and a knowledge-aware attention to mine tourists’ long-term interest from the KG. And a personalized attention-based GRU network is employed to extract insights from recent travel records for short-term interest modeling. These two interests are adaptively combined through a gated fusion network to generate personalized recommendations. Our experiments on real-world datasets demonstrated the effectiveness of KGNARec, achieving an 8.6 % improvement in AUC and a 7.1 % improvement in ACC compared to strongest baselines. Moreover, parameter sensitivity analysis validated the robustness of the model design, while ablation studies highlighted the rationality and contributions of each individual component. Finally, a case study showcased the superior interpretability of KGNARec, particularly in uncovering tourists’ decision-making processes. These features provide destination marketing organizations with actionable insights to optimize their services and marketing strategies.

Despite these promising results, our study has certain limitations that warrant further investigation. Firstly, the model primarily focuses on individual behavioral patterns and does not explicitly account for collaborative effects or social influences. This limitation could result in the omission of higher-order collaborative interactions. Future work should consider integrating physical constraints and social awareness to improve recommendation performance. Secondly, our model employs a shallow GCN for knowledge extraction, which may restrict its capacity to capture more comprehensive tourism knowledge. While deeper architectures introduce challenges such as overfitting and increased computational complexity, future efforts will focus on developing deeper yet lightweight GCNs. Finally, the reliance on a static tourism KG prevents the model from incorporating real-time updates and changes in tourism knowledge. Future research will explore temporal knowledge graphs or multi-modal knowledge graphs, enabling the model to reflect real-time trends and dynamic market changes. In summary, the KG, as a powerful tool, heralds boundless innovation potential and broad application prospects in the field of smart tourism.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Abbasi-Moud, Z., Vahdat-Nejad, H., & Sadri, J. (2021). Tourism recommendation system based on semantic clustering and sentiment analysis. *Expert Systems with Applications*, 167, Article 114324.
- Aghaei, S., Raad, E., & Fensel, A. (2022). Question answering over knowledge graphs: a case study in Tourism. *IEEE Access*, 10, 1.
- Ameen, T., Chen, L., Xu, Z., Lyu, D., & Shi, H. (2020). A convolutional neural network and matrix factorization-based travel location recommendation method using community-contributed geotagged photos. *ISPRS International Journal of Geo-Information*, 9(8), 464.
- Batmaz, Z., Yurekli, A., Bilge, A., & Kaleli, C. (2019). A review on deep learning for recommender systems: Challenges and remedies. *Artificial Intelligence Review*, 52(1), 1–37.
- Cai, L., Xu, J., Liu, J., & Pei, T. (2018). Integrating spatial and temporal contexts into a factorization model for POI recommendation. *International Journal of Geographical Information Science*, 32(3), 524–546.
- Cai, X., Xie, L., Tian, R., & Cui, Z. (2022). Explicable recommendation based on knowledge graph. *Expert Systems with Applications*, 200, Article 117035.
- Cao, G., Cui, S., & Joe, I. (2023). Improving the spatial-temporal aware attention network with dynamic trajectory graph learning for next Point-Of-Interest recommendation. *Information Processing & Management*, 60(3), Article 103335.
- Chaudhari, K., & Thakkar, A. (2020). A comprehensive survey on travel recommender systems. *Archives of Computational Methods in Engineering*, 27(5), 1545–1571.
- Chen, L., Cao, J., Wang, Y., Liang, W., & Zhu, G. (2022). Multi-view graph attention network for travel recommendation. *Expert Systems with Applications*, 191, Article 116234.
- Chen, L., Cao, J., Zhu, G., Wang, Y., & Liang, W. (2021). A multi-task learning approach for improving travel recommendation with keywords generation. *Knowledge-Based Systems*, 233, Article 107521.
- Chen, L., Lyu, D., Yu, S., & Chen, G. (2023). Multi-level visual similarity based personalized tourist attraction recommendation using geo-tagged photos. *ACM Transactions on Knowledge Discovery from Data*, 17(7), 1–18.
- Chen, X., Jia, S., & Xiang, Y. (2020). A review: Knowledge reasoning over knowledge graph. *Expert Systems with Applications*, 141, Article 112948.
- Chicaiza, J., & Valdiviezo-Diaz, P. (2021). A comprehensive survey of knowledge graph-based recommender systems: Technologies, development, and contributions. *Information*, 12(6), 232.
- Cui, G., Luo, J., & Wang, X. (2018). Personalized travel route recommendation using collaborative filtering based on GPS trajectories. *International Journal of Digital Earth*, 11(3), 284–307.
- Dai, Q., Wu, X., Fan, L., Li, Q., Liu, H., Zhang, X., Wang, D., Lin, G., & Yang, K. (2022). Personalized knowledge-aware recommendation with collaborative and attentive graph convolutional networks. *Pattern Recognition*, 128, Article 108628.
- Deng, N., Liu, J., Dai, Y., & Li, H. (2019). Different cultures, different photos: A comparison of Shanghai's pictorial destination image between East and West. *Tourism Management Perspectives*, 30, 182–192.
- Feng, J., Li, Y., Zhang, C., Sun, F., Meng, F., Guo, A., & Jin, D. (2018). *DeepMove: Predicting Human Mobility with Attentional Recurrent Networks*. Paper presented at the Proceedings of the 2018 World Wide Web Conference, Republic and Canton of Geneva, Switzerland, 1459–1468.
- Gao, J., Peng, P., Lu, F., Claramunt, C., Qiu, P., & Xu, Y. (2024). Mining tourist preferences and decision support via tourism-oriented knowledge graph. *Information Processing & Management*, 61(1), Article 103523.
- Gao, J., Peng, P., Lu, F., Claramunt, C., & Xu, Y. (2023). Towards travel recommendation interpretability: Disentangling tourist decision-making process via knowledge graph. *Information Processing & Management*, 60(4), Article 103369.
- Gao, Y., Gao, X., Li, X., Yao, B., & Chen, G. (2022). An embedded GRASP-VNS based two-layer framework for tour recommendation. *IEEE Transactions on Services Computing*, 15(2), 847–859.
- Guo, Q., Zhuang, F., Qin, C., Zhu, H., Xie, X., Xiong, H., & He, Q. (2020). A survey on knowledge graph-based recommender systems. *IEEE Transactions On Knowledge and Data Engineering*, 34(8), 3549–3568.
- Hamid, R. A., Albahri, A. S., Alwan, J. K., Al-qaysi, Z. T., Albahri, O. S., Zaidan, A. A., Alnoor, A., Alamoodi, A. H., & Zaidan, B. B. (2021). How smart is e-tourism? A systematic review of smart tourism recommendation system applying data management. *Computer Science Review*, 39, Article 100337.
- He, S., Du, W., Zhang, Y., Chen, L., Chen, Z., & Chen, N. (2024). Next location prediction using heterogeneous graph-based fusion network with physical and social awareness. *International Journal of Geographical Information Science, IJGIS*, 1–26.
- He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T. (2017). Neural Collaborative Filtering. *Proceedings of the 26th international conference on world wide web* (pp. 173–182). Perth, Australia.
- Henar Salas-Olmedo, M., Moya-Gomez, B., Carlos Garcia-Palomares, J., & Gutierrez, J. (2018). Tourists' digital footprint in cities: Comparing Big Data sources. *Tourism Management*, 66, 13–25.
- Hu, B., Shi, C., Zhao, W., & Yu, P. S. (2018). *Leveraging meta-path based context for top-N recommendation with a neural co-attention model*. Paper presented at the Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, London, England, 1531–1540.
- Hu, L., Li, C., Shi, C., Yang, C., & Shao, C. (2020). Graph neural news recommendation with long-term and short-term interest modeling. *Information Processing & Management*, 57(2), Article 102142.
- Huang, Y., & Bian, L. (2009). A Bayesian network and analytic hierarchy process based personalized recommendations for tourist attractions over the Internet. *Expert Systems with Applications*, 36(1), 933–943.
- Jannach, D., Manzoor, A., Cai, W., & Chen, L. (2022). A Survey on Conversational Recommender Systems. *ACM Computing Surveys*, 54(5), 1–36.
- Ji, S., Pan, S., Cambria, E., Marttinen, P., & Yu, P. S. (2022). A survey on knowledge graphs: representation, acquisition, and applications. *IEEE Transaction on Neural Networks and Learning Systems*, 33(2), 494–514.
- Jiang, S., Qian, X., Shen, J., Fu, Y., & Mei, T. (2015). author topic model based collaborative filtering for personalized POI recommendation. *IEEE Transactions on Multimedia*, 17(6), 1.
- Jin, C., Cheng, J., & Xu, J. (2018). Using user-generated content to explore the temporal heterogeneity in tourist mobility. *Journal of Travel Research*, 57(6), 779–791.
- Khan, N., Ma, Z., Ullah, A., & Polat, K. (2022). Categorization of knowledge graph based recommendation methods and benchmark datasets from the perspectives of application scenarios: A comprehensive survey. *Expert Systems with Applications*, 206, Article 117737.
- Law, R., Li, G., Fong, D. K. C., & Han, X. (2019). Tourism demand forecasting: A deep learning approach. *Annals of Tourism Research*, 75, 410–423.
- Li, M., Ma, W., & Chu, Z. (2024). KGIE: Knowledge graph convolutional network for recommender system with interactive embedding. *Knowledge-Based Systems*, 295, Article 111813.
- Li, M., Sun, Z., Zhang, S., & Zhang, W. (2021). Enhancing knowledge graph embedding with relational constraints. *Neurocomputing*, 429, 77–88.
- Li, X., Hu, R., & Wang, Z. (2022). Next-point-of-interest recommendation based on joint mining of regularity and randomness. *Knowledge-Based Systems*, 258, Article 110052.
- Li, Y., Hu, C., Huang, C., & Duan, L. (2017). The concept of smart tourism in the context of tourism information services. *Tourism Management*, 58, 293–300.
- Liu, J., Chen, Y., Huang, X., Li, J., & Min, G. (2023). GNN-based long and short term preference modeling for next-location prediction. *Information Sciences*, 629, 1–14.
- Liu, Y., Yang, S., Xu, Y., Miao, C., Wu, M., & Zhang, J. (2023). Contextualized graph attention network for recommendation with item knowledge graph. *IEEE Transactions on Knowledge and Data Engineering*, 35(1), 181–195.
- Liu, Z., Zhang, D., Zhang, C., Bian, J., Deng, J., Shen, G., & Kong, X. (2023). KDRank: Knowledge-driven user-aware POI recommendation. *Knowledge-Based Systems*, 278, Article 110884.
- May Petry, L., Leite Da Silva, C., Esuli, A., Renso, C., & Bogorny, V. (2020). MARC: A robust method for multiple-aspect trajectory classification via space, time, and semantic embeddings. *International Journal of Geographical Information Science*, 34 (7), 1428–1450.
- Mezni, H., Benslimane, D., & Bellatreche, L. (2021). Context-aware service recommendation based on knowledge graph embedding. *IEEE Transactions on Knowledge and Data Engineering*, 1.
- Mou, N., Jiang, Q., Zhang, L., Niu, J., Zheng, Y., Wang, Y., & Yang, T. (2022). Personalized tourist route recommendation model with a trajectory understanding via neural networks. *International Journal of Digital Earth*, 15(1), 1738–1759.
- Mou, N., Zheng, Y., Makkonen, T., Yang, T., Tang, J. J., & Song, Y. (2020). Tourists' digital footprint: The spatial patterns of tourist flows in Qingdao. *China. Tourism Management*, 81, Article 104151.
- Renjith, S., Sreekumar, A., & Jathavedan, M. (2020). An extensive study on the evolution of context-aware personalized travel recommender systems. *Information Processing & Management*, 57(1), Article 102078.
- Rossi, A., Barbosa, D., Firmani, D., Matinani, A., & Merialdo, P. (2021). Knowledge graph embedding for link prediction: A comparative analysis. *ACM Transactions on Knowledge Discovery from Data*, 15(2), 1–49.
- Sang, L., Xu, M., Qian, S., & Wu, X. (2021). Knowledge graph enhanced neural collaborative recommendation. *Expert Systems with Applications*, 164, Article 113992.
- Shi, W., Liu, Z., An, Z., & Chen, P. (2021). RegNet: A neural network model for predicting regional desirability with VGI data. *International Journal of Geographical Information Science*, 35(1), 175–192.
- Shokrzadeh, Z., Feizi-Derakhshi, M., Balafar, M., & Bagherzadeh Mohasefi, J. (2024). Knowledge graph-based recommendation system enhanced by neural collaborative filtering and knowledge graph embedding. *Ain Shams Engineering Journal*, 15(1), Article 102263.
- Sperli, G. (2021). A cultural heritage framework using a Deep Learning based Chatbot for supporting tourist journey. *Expert Systems with Applications*, 183, Article 115277.
- Sun, Y., Fan, H., Bakillah, M., & Zipf, A. (2015). Road-based travel recommendation using geo-tagged images. *Computers, Environment and Urban Systems*, 53, 110–122.

- Wan, L., Hong, Y., Huang, Z., Peng, X., & Li, R. (2018). A hybrid ensemble learning method for tourist route recommendations based on geo-tagged social networks. *International Journal of Geographical Information Science*, 32(11), 2225–2246.
- Wang, C., Yuan, M., Yang, Y., Peng, K., & Jiang, H. (2024). Revisiting long- and short-term preference learning for next POI recommendation With hierarchical LSTM. *IEEE Transactions on Mobile Computing*, 23(12), 12693–12705.
- Wang, F., Li, Y., Zhang, Y., & Wei, D. (2022). KLGcn: Knowledge graph-aware Light Graph Convolutional Network for recommender systems. *Expert Systems with Applications*, 195, Article 116513.
- Wang, F., Zheng, Z., Zhang, Y., Li, Y., Yang, K., & Zhu, C. (2023). To see further: Knowledge graph-aware deep graph convolutional network for recommender systems. *Information Sciences*, 647, Article 119465.
- Wang, H., Zhang, F., Wang, J., Zhao, M., Li, W., Xie, X., & Guo, M. (2019). Exploring high-order user preference on the knowledge graph for recommender systems. *ACM Transactions on Information Systems*, 37(3), 1–26.
- Wang, H., Zhao, M., Xie, X., Li, W., & Guo, M. (2019). Knowledge graph convolutional networks for recommender systems. Paper presented at the Proceedings of the World Wide Web conference (WWW 2019), San Francisco, CA, 3307–3313.
- Wang, K., Wang, X., & Lu, X. (2023). POI recommendation method using LSTM-attention in LBSN considering privacy protection. *Complex & Intelligent Systems*, 9(3), 2801–2812.
- Wang, W., Shen, X., Yi, B., Zhang, H., Liu, J., & Dai, C. (2024). Knowledge-aware fine-grained attention networks with refined knowledge graph embedding for personalized recommendation. *Expert Systems with Applications*, 249, Article 123710.
- Wu, L., He, X., Wang, X., Zhang, K., & Wang, M. (2023). A Survey on Accuracy-Oriented Neural Recommendation: from Collaborative Filtering to Information-Rich Recommendation. *IEEE Transactions on Knowledge and Data Engineering*, 35(5), 4425–4445.
- Wu, S., Sun, F., Zhang, W., Xie, X., & Cui, B. (2023). Graph neural networks in recommender systems: a survey. *ACM Computing Surveys*, 55(5), 1–37.
- Wu, Y., Li, K., Zhao, G., & Qian, X. (2022). Personalized long- and short-term preference learning for next poi recommendation. *IEEE Transactions On Knowledge and Data Engineering*, 34(4), 1944–1957.
- Xiaoyu Sun, Z. H. X. P. (2018). Building a model-based personalized recommendation approach for tourist attractions from geotagged social media data. *International Journal of Digital Earth*, 12(6), 661–678.
- Xie, F., Zheng, A., Chen, L., & Zheng, Z. (2021). Attentive Meta-graph Embedding for item Recommendation in heterogeneous information networks. *Knowledge-Based Systems*, 211, Article 106524.
- Xu, Z., Chen, L., Dai, Y., & Chen, G. (2017). A dynamic topic model and matrix factorization-based travel recommendation method exploiting ubiquitous data. *IEEE Transactions on Multimedia*, 19(8), 1933–1945.
- Yang, Z., & Dong, S. (2020). HAGERec: Hierarchical attention graph convolutional network incorporating knowledge graph for explainable recommendation. *Knowledge-Based Systems*, 204, Article 106194.
- Yao, Y., Guo, Z., Dou, C., Jia, M., Hong, Y., Guan, Q., & Luo, P. (2023). Predicting mobile users' next location using the semantically enriched geo-embedding model and the multilayer attention mechanism. *Computers, Environment and Urban Systems*, 104, Article 102009.
- Yi, B., Zhou, H., Linna, L., Yaoli, W., & Yu, L. (2020). A BiLSTM-CNN model for predicting users' next locations based on geotagged social media. *International Journal of Geographical Information Science*, 35(4), 639–660.
- Zhang, F., Yuan, N. J., Lian, D., Xie, X., & Ma, W. (2016). Collaborative knowledge base embedding for recommender systems. Paper presented at the Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, San Francisco, CA, 353–362.
- Zhang, G., Li, H., Li, S., Wang, B., & Ding, Z. (2024). MMKG-PAR: multi-modal knowledge graphs-based personalized attraction recommendation. *Sustainability*, 16(5), 2211.
- Zhang, J., Liu, X., Zhou, X., & Chu, X. (2021). Leveraging graph neural networks for point-of-interest recommendations. *Neurocomputing*, 462, 1–13.
- Zhang, S., Yao, L., Sun, A., & Tay, Y. (2019). Deep learning based recommender system: A survey and new perspectives. *ACM Computing Surveys*, 52(1), 1–38.
- Zheng, W., Huang, X., & Li, Y. (2017). Understanding the tourist mobility using GPS: Where is the next place? *Tourism Management*, 59, 267–280.
- Zheng, W., Liao, Z., & Qin, J. (2017). Using a four-step heuristic algorithm to design personalized day tour route within a tourist attraction. *Tourism Management*, 62, 335–349.
- Zheng, Y., Mou, N., Zhang, L., Makkonen, T., & Yang, T. (2021). Chinese tourists in Nordic countries: An analysis of spatio-temporal behavior using geo-located travel blog data. *Computers, Environment and Urban Systems*, 85, Article 101561.
- Zhu, G., Cao, J., Chen, L., Wang, Y., Bu, Z., Yang, S., Wu, J., & Wang, Z. (2023). A multi-task graph neural network with variational graph auto-encoders for session-based travel packages recommendation. *Acm Transactions On the Web*, 17(3), 1–30.
- Zhu, G., Wang, Y., Cao, J., Bu, Z., Yang, S., Liang, W., & Liu, J. (2021). Neural attentive travel package recommendation via exploiting long-term and short-term behaviors. *Knowledge-Based Systems*, 211, Article 106511.